

An Ensemble of Robust Detectors for Crypto Shock Alerts: A Pre-Registered Evaluation on 40 Stress Days

SENTINEL project report · October 2026 · Code and protocol: github.com/nathencoelhoo/sentinel

Abstract. We evaluate SENTINEL, a streaming anomaly detector that combines five robust statistical detectors (median/MAD z-score, EWMA volatility ratio, CUSUM, Bayesian online changepoint detection, volume spike) through a logistic ensemble with per-symbol adaptive alert thresholds. The question was whether the ensemble flags crypto market shocks earlier, or with fewer false alarms, than a z-score baseline at the same nominal alarm budget. The protocol was frozen before the main run: the 40 highest-range days for BTCUSDT and ETHUSDT since 2020, a price-only onset rule, a random-alert chance baseline, and cluster-bootstrap intervals. At a nominal 6 alerts/day, SENTINEL emitted 1.9 fewer alerts per quiet day than an adaptive z-score baseline (4.8 vs 6.7; paired 95% interval -2.3 to -1.4) with no clear change in event recall (-0.03 [-0.14 , $+0.08$]). We found no clear evidence of earlier detection, no clear benefit from online-learned weights, and no single detector whose removal clearly changed recall or latency. Recall above a chance alerter was small ($+0.06$) and is not established. The data support a narrow claim about alert volume, not about detection skill.

1. Introduction

Streaming anomaly detectors for markets are easy to build and hard to evaluate. Volatility clusters, events are rare and hand-labelled, and a detector that alarms often will "catch" almost any event. We built SENTINEL, an open-source streaming detector for crypto markets, and asked one narrow question: at the same nominal alarm budget, does an ensemble of robust detectors flag large shocks earlier, or with fewer false alarms, than a single z-score? This report documents the system, a protocol fixed before the main evaluation ran, and a result that is more modest than the system's ambition. Contributions: (i) a causal streaming implementation shared by live and replay use, with 57 automated tests, including one asserting that two engines fed different futures produce identical past outputs; (ii) an evaluation with a random-alert chance baseline and cluster-bootstrap intervals; (iii) an explicit account of what the evidence does and does not support.

2. System

2.1 Streaming design. Trades are aggregated into 1-minute bars. Every component is a causal state machine that sees bar t only after bars $\leq t$. The same code runs live (in a Web Worker) and in replay, so there is no train/serve skew.

2.2 Detectors. Let r_t be the 1-minute log return. (a) *Robust z*: $z_t = |r_t - \text{med}| / (1.4826 \cdot \text{MAD})$ over the preceding 240 returns [3]. (b) *EWMA volatility ratio*: $\sigma_{\text{fast}} / \sigma_{\text{slow}}$ with decay 0.94 and 0.995 [4]. (c) *CUSUM* [2] on standardised returns: $S_t^+ = \max(0, S_{t-1}^+ + z_t - k)$, S_t^- symmetric, $k = 0.5$, restart at $h = 10$. (d) *Bayesian online changepoint detection* [1] with a Normal-Gamma model and constant hazard $1/200$; the statistic is the posterior mass on run lengths below 5. (e) *Volume spike*: v_t divided by the trailing 240-bar median. An order-book-imbalance detector exists in the code but is unused here, because free historical level-2 data do not exist.

2.3 Common scale and ensemble. Each raw statistic s_j is mapped through its own rolling empirical CDF (2,000 observations, queried before insertion) to u_j , then to a surprise feature $x_j = -\ln(1 - u_j)$, which is $\text{Exp}(1)$ under stationarity. The ensemble is logistic, $p = \sigma(b + \sum_j w_j x_j)$. The *fixed* variant uses the prior $w_j = 1.5/n$, $b = -4.5$, i.e. $p = \sigma(1.5(\bar{x} - 3))$. The *learned* variant runs online SGD on log-loss with an L2 pull toward that prior, non-negative weights, and delayed weak labels (a return of at least $4\hat{\sigma}$ in the next 30 bars, applied only after those 30 bars have elapsed, so no label uses information the engine could not have had).

2.4 Alerting. The threshold is the $(1 - \rho)$ quantile of the previous 1,440 scores, with $\rho = (\text{alarms/day} \times 60 \text{ s})/86,400$ set for 6 alarms/day, plus a 15-bar cooldown that merges clustered exceedances. A severity-escalation rule lets an alert through inside a cooldown if its score beats the $(1 - \rho/10)$ quantile and the previous alert's score; it was added after a synthetic test showed two false alarms masking a real jump.

2.5 Baselines. *zscore_fixed3*: $|z| > 3$ with a 100-bar rolling mean and standard deviation. *zscore_matched* (the reference): the same statistic with exactly SENTINEL's adaptive threshold, cooldown and escalation, which isolates what the detectors and ensemble add beyond thresholding.

3. Evaluation protocol

Events. Binance Vision 1-minute spot klines [6] for BTCUSDT and ETHUSDT. Events come from a price-only rule: rank UTC days from 2020-01-01 to 2026-08-31 by $(\text{high} - \text{low})/\text{open}$, take the top 20 per symbol, skipping any day within three days of a chosen one. This gave 40 events; none were excluded. **Onset and detection.** Onset is the first minute the close is at least 1% from the UTC day's open, identically for all methods. An event is detected if an alert falls in $[\text{onset} - 30, \text{onset} + 120]$ minutes; latency is the first such alert minus onset (negative

= early warning). **Quiet periods.** After a 3-day warm-up, quiet time excludes every selected event day of the same symbol from 60 minutes before to 12 hours after. Quiet-period alerts per day are therefore an *upper bound* on false alarms, since other volatile periods remain. **Chance baseline.** For each method, a random alerter with the same quiet-period alert rate and minimum spacing is simulated 1,000 times; "recall minus chance" removes the advantage of simply alerting more. **Inference.** 95% percentile intervals from a cluster bootstrap over calendar days (BTC and ETH events on the same day share a cluster), 2,000 resamples [5]; differences against the reference use identical resamples (paired). No multiplicity correction is applied; we read intervals, not p-values.

Pre-registration. All parameters live in one file, hashed (SHA-256 prefix c13175749b1c) and committed (f07a4118bb) before event selection ran, then executed by a GitHub Actions workflow. This is self-hosted registration, not a third-party registry. A five-event pilot replay (COVID crash, Terra/Luna, FTX) preceded the study and motivated its design; no parameter was changed afterwards.

4. Results

Method	Recall [95% CI]	Chance recall [95% range]	Recall – chance	Quiet alerts/day [95% CI]	Median latency, min [95% CI]
zscore_fixed3	0.82 [0.69, 0.93]	0.82 [0.70, 0.93]	+0.01	14.11 [13.38, 14.87]	0 [–3, 3]
zscore_matched (ref.)	0.50 [0.33, 0.66]	0.53 [0.38, 0.68]	–0.03	6.65 [6.24, 7.10]	2 [–3, 9]
sentinel_fixed	0.47 [0.30, 0.64]	0.41 [0.28, 0.55]	+0.06	4.79 [4.52, 5.07]	0 [–3, 49]
sentinel_learned	0.45 [0.28, 0.62]	0.39 [0.25, 0.55]	+0.06	4.60 [3.99, 5.20]	7 [–3, 46]
– robust_z	0.45 [0.28, 0.62]	0.39 [0.25, 0.55]	+0.06	4.47 [4.18, 4.78]	–1 [–3, 45]
– ewma_vol	0.55 [0.38, 0.71]	0.44 [0.30, 0.60]	+0.11	5.25 [4.94, 5.60]	0 [–3, 22]
– cusum	0.53 [0.35, 0.70]	0.42 [0.28, 0.57]	+0.11	4.90 [4.56, 5.23]	1 [–2, 22]
– bocpd	0.42 [0.26, 0.59]	0.41 [0.28, 0.55]	+0.02	4.74 [4.33, 5.14]	0 [–3, 51]
– volume_spike	0.47 [0.31, 0.64]	0.42 [0.27, 0.57]	+0.05	4.92 [4.64, 5.22]	0 [–3, 27]

Table 1. Per-method results on 40 events. Ablation rows remove one detector from the fixed ensemble. "Quiet alerts/day" is the alert rate outside event periods. Chance recall is for a random alerter at the method's own alert rate.

Method	Recall difference	Quiet alerts/day difference	Latency difference, min (events both detect)
zscore_fixed3	+0.32 [0.16, 0.50]	+7.46 [6.62, 8.28]	0 [–6, 0]
sentinel_fixed	–0.03 [–0.14, 0.08]	–1.86 [–2.31, –1.42]	0 [0, 1]
sentinel_learned	–0.05 [–0.19, 0.09]	–2.06 [–2.65, –1.45]	1 [0, 14]
– robust_z	–0.05 [–0.20, 0.10]	–2.18 [–2.66, –1.68]	0 [0, 22]
– ewma_vol	+0.05 [–0.07, 0.16]	–1.40 [–1.91, –0.90]	0 [0, 0]
– cusum	+0.03 [–0.09, 0.13]	–1.76 [–2.24, –1.30]	0 [0, 1]
– bocpd	–0.08 [–0.21, 0.05]	–1.92 [–2.43, –1.37]	0 [0, 19]
– volume_spike	–0.03 [–0.13, 0.08]	–1.73 [–2.22, –1.26]	0 [0, 1]

Table 2. Paired differences versus zscore_matched (identical bootstrap resamples). **Bold** = 95% interval excludes 0.

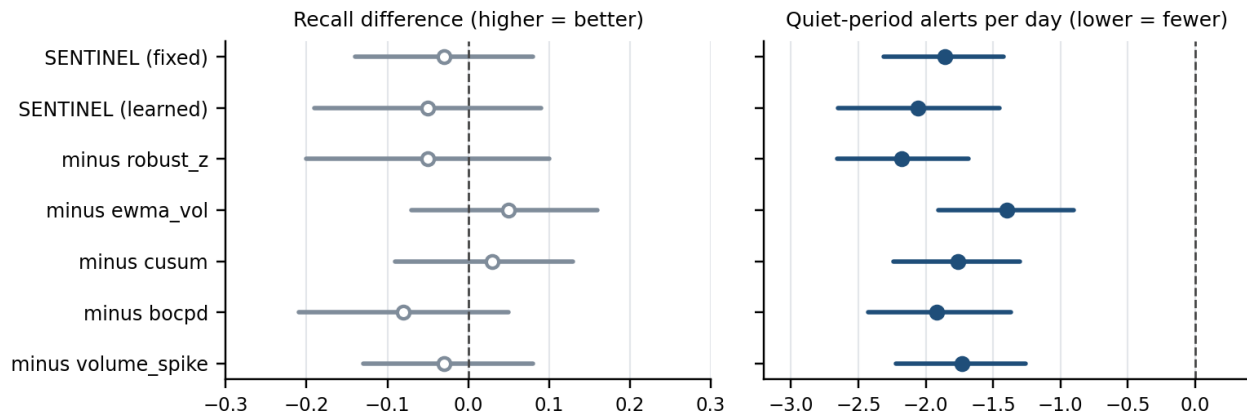


Figure 1. Paired differences versus zscore_matched. Filled markers: interval excludes 0; hollow: it does not.

F1. Alert volume fell, clearly. SENTINEL (fixed) emitted 4.79 quiet-period alerts/day versus 6.65 for the matched baseline; the paired difference is $-1.86 [-2.31, -1.42]$. Every ablated variant shows a similar reduction (-1.4 to -2.2), which suggests the reduction comes from the shared pipeline (ECDF normalisation, ensembling, thresholding) rather than from any one detector. That is a hypothesis consistent with the data, not a tested mechanism.

F2. Recall did not clearly change. $-0.03 [-0.14, +0.08]$ for the fixed ensemble; every ablation interval also includes 0.

F3. The classic rule has no skill. $|z| > 3$ reached recall 0.82, exactly what a random alerter achieves at its 14.1 alerts/day. High recall there is bought with alert volume.

F4. No evidence of earlier detection. Median latencies are 0 to 7 minutes with wide intervals (e.g. 0 $[-3, 49]$); every paired latency interval includes 0.

F5. Learning and components. The learned ensemble is no better than the fixed one. Removing the EWMA or CUSUM detector gave the highest recall (0.55, 0.53) but also more alerts (5.25, 4.90/day), a pattern consistent with a volume-recall trade-off rather than a detector effect; every such difference is inside the noise.

F6. Chance-adjusted skill is unproven. SENTINEL's recall exceeds its chance level by 0.06, the matched z-score's by -0.03 , the classic rule's by 0.01. The recall and chance intervals overlap widely, and we did not compute a paired interval for skill, so we do not claim it.

5. Limitations and threats to validity

- **Nominal versus realised alert rates.** The same nominal budget (6/day) produced different realised rates (4.8 vs 6.7). A lower alert rate mechanically lowers the chance of a hit, so F1 and F2 together are not a like-for-like comparison. The chance adjustment helps but is incomplete; a fair test needs recall-versus-alert-rate curves or budgets matched on realised rate.
- **Onset rule.** In 22 of 40 events (55%; marked † in Appendix A) the onset falls within 30 minutes after 00:00 UTC. These shocks began before the day opened (the March 2020 crash straddles midnight), so a day-open anchor misdates them and adds noise to recall and latency for every method.
- **Adaptive-threshold self-inflation.** Because the threshold is a one-day quantile, a prolonged crisis raises its own threshold; this may lower recall on later days of multi-day stress (March 2020, May 2021). We did not test this.
- **Selection on the outcome.** Events are the largest-range days, so results speak to obvious shocks, not to the subtle regime changes an ensemble is meant to catch.
- **Sample size.** Forty events, strongly correlated across BTC and ETH, and no multiplicity correction. Ablations are compared with the reference, not directly with the full ensemble.
- **Cold starts.** Each event uses a fresh engine and 3 days of warm-up, which handicaps the learned variant.
- **Registration and scope.** The pre-registration is a hashed commit in the author's own repository. No trading performance is claimed. Data are CC BY-NC-SA 4.0 (non-commercial); this is research, not advice.

6. Conclusion and next steps

At a nominal 6 alerts/day, an ensemble of robust detectors with adaptive thresholds produced clearly fewer alerts than an equally thresholded z-score, with no clear loss of recall on 40 large crypto shocks. We found no evidence that it detects them earlier, and the chance-adjusted skill is unproven. The follow-up we propose, labelled exploratory because it is designed after seeing these results: (1) a trailing-reference onset rule that does not depend on midnight; (2) recall-versus-realised-alert-rate curves and paired skill intervals; (3) a held-out set (the next 40 ranked days) plus subtle events (volatility regime shifts, drifts) where an ensemble should matter; (4) longer warm-up for the learned variant; (5) recording live level-2 data to test the order-book detector.

References

- [1] Adams, R. P. and MacKay, D. J. C. (2007). Bayesian online changepoint detection. arXiv:0710.3742.
- [2] Page, E. S. (1954). Continuous inspection schemes. *Biometrika* 41(1/2), 100–115.
- [3] Rousseeuw, P. J. and Croux, C. (1993). Alternatives to the median absolute deviation. *J. Amer. Statist. Assoc.* 88(424), 1273–1283.
- [4] J.P. Morgan/Reuters (1996). *RiskMetrics Technical Document*, 4th ed.
- [5] Efron, B. (1979). Bootstrap methods: another look at the jackknife. *Ann. Statist.* 7(1), 1–26.
- [6] Binance Vision public data, data.binance.vision (licence CC BY-NC-SA 4.0).

Appendix A. The 40 events

Selected by the pre-registered rule (top 20 daily ranges per symbol, at least 3 days apart). Onset = first minute the close is at least 1% from the UTC day's open. † = onset within 30 minutes after 00:00 UTC (22 of 40). Range % = (high – low)/open for the UTC day.

BTCUSDT day	Onset (UTC)	Range %	ETHUSDT day	Onset (UTC)	Range %
2020-03-13 †	00:00	45.3	2020-03-13 †	00:00	49.9
2021-05-19 †	00:07	31.7	2021-05-19 †	00:07	46.0
2021-02-08	02:02	22.7	2021-01-03	05:03	31.3
2021-12-04	00:48	22.1	2021-05-23 †	00:02	28.4
2020-03-19	02:04	21.3	2025-02-03 †	00:16	27.7
2021-01-11 †	00:04	20.6	2021-01-11 †	00:04	27.5
2021-01-29 †	00:05	19.8	2020-03-20 †	00:02	26.3
2021-09-07	07:11	19.1	2021-02-23 †	00:05	24.1
2021-05-23 †	00:01	19.1	2021-09-07	07:12	24.0
2022-06-13	00:59	18.7	2022-11-10 †	00:02	23.9
2026-02-06 †	00:05	18.7	2020-08-02 †	00:08	23.5
2022-02-28	01:20	18.0	2025-05-08	01:00	23.0
2020-03-16 †	00:01	17.2	2020-09-05 †	00:22	22.5
2021-02-22 †	00:13	17.2	2022-06-19 †	00:10	22.4
2022-11-08	03:18	17.2	2026-08-19	12:47	22.3
2025-10-10	15:00	16.9	2021-01-21 †	00:15	22.1
2021-05-12	00:37	16.6	2025-10-10	06:52	21.9
2021-01-04	00:31	16.6	2024-08-05 †	00:12	21.8
2023-10-23	01:26	16.2	2024-05-20	04:34	21.0
2021-01-22 †	00:03	16.1	2022-01-21	00:41	21.0

Appendix B. Reproducing the study

Everything runs on GitHub Actions with no local setup. The workflow named *study* installs the project, runs the 57 tests, applies the event rule to Binance Vision daily klines, downloads the 1-minute files for the 40 events, replays every method, and writes the tables above (results/study/summary.md and summary.json, which also record the config hash and commit). The frozen parameters are in eval/study.json; the protocol is described in docs/STUDY.md.